

1. How do you run a process in the background in Linux?

Add & at the end of a command to run it in the background.

Example: python app.py &

2. Write a simple shell script to automate backup creation.

```
#!/bin/bash
```

```
cp -r /home/user/Documents /home/user/Backup  
echo "Backup created successfully."
```

3. How do you create, copy, move, and delete files in Linux?

- Create a file: touch file.txt
- Copy a file: cp file.txt backup.txt
- Move or rename a file: mv file.txt newfile.txt
- Delete a file: rm file.txt

4. Write a shell script to perform string manipulation.

```
#!/bin/bash
```

```
echo "Enter a string:"  
read str
```

```
echo "Uppercase: ${str^^}"  
echo "Length: ${#str}"
```

5. How do you compress and extract files using tar and gzip?

- Compress:

```
tar -czvf backup.tar.gz folder/
```

- Extract:

```
tar -xzvf backup.tar.gz
```

6. Explain command substitution and process substitution with examples.

Command substitution: It stores the output of a command in a variable.

Example: `today=$(date)`
`echo $today`

Process substitution: It uses the output of a command as input to another command.

Example: `diff <(ls dir1) <(ls dir2)`